

9(a)(i)	A field of force is a region of space where a body possessing a particular property experiences a force due to the presence of other bodies with the same property, without any physical contact between them.
(a)(ii)1.	Gravitational field
(a)(ii)2.	Electric field and gravitational field
(b)(i)	Time taken $t = \frac{0.12}{6.7 \times 10^7} = 1.8 \times 10^{-9} \text{ s}$
(b)(ii)	Electric field strength $E = \frac{V}{d} = \frac{960}{0.024} = 4.0 \times 10^4 \text{ N C}^{-1}$ Vertical acceleration $a = \frac{F}{m} = \frac{qE}{m} = \frac{(1.6 \times 10^{-19}) 40000}{(9.11 \times 10^{-31})} = 7.0 \times 10^{15} \text{ m s}^{-2}$
(c)	Vertical displacement $= u_y t + \frac{1}{2} a_y t^2$ $= 0 + \frac{1}{2} (7.03 \times 10^{15}) (1.79 \times 10^{-9})^2$ $= 0.0113 \text{ m}$ Since plate A is 0.012 m (greater than 0.0113 m) above the initial point of entry of electron, the electron will not collide with the metal plate.
(d)(i)	By Fleming's left hand rule, the magnetic force is always perpendicular to the direction of motion of the particle and the magnetic field. As a result, the magnetic force will only cause a change in direction of motion and not the magnitude of its velocity, and thus causing the magnitude of the magnetic force will remain unchanged. This magnetic force is always pointing towards a fixed point being the centre of the circle. The magnetic force thus provides the centripetal force required for the particle to move in a circular path.

(d)(ii)	$Bqv = \frac{mv^2}{r}$ $m = \frac{Bqr}{v}$ $= \frac{(0.090)(1.60 \times 10^{-19})(0.074)}{4.6 \times 10^4}$ $= 2.317 \times 10^{-26} \text{ kg}$ $= \frac{2.317 \times 10^{-26}}{1.66 \times 10^{-27}}$ $= 14 \text{ u}$
(e)(i)	$Bqv = \frac{mv^2}{r}$ $r = \frac{mv}{Bq}$ All else being constant, a particle of smaller mass will have a smaller radius of curvature.
(e)(ii)	$Bqv = \frac{mv^2}{r}$ $r = \frac{mv}{Bq}$ All else being constant, a particle with double the charge will have half the radius. Diameter will be 7.4 cm.